



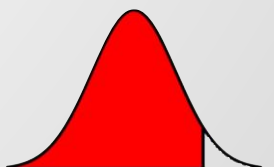
Imbalance-Aware Feature Selection and Machine Learning for U.S. Corporate Bankruptcy Prediction

Yun-Chen Li, Rou-Rou Yu, Yi-An Chen, Huei-Wen Tung



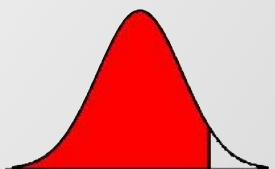
Outline

1. Motivation
2. Data (EDA)
3. Methodology (Benchmark)



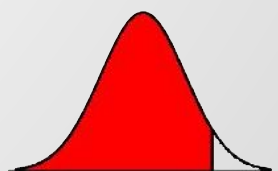
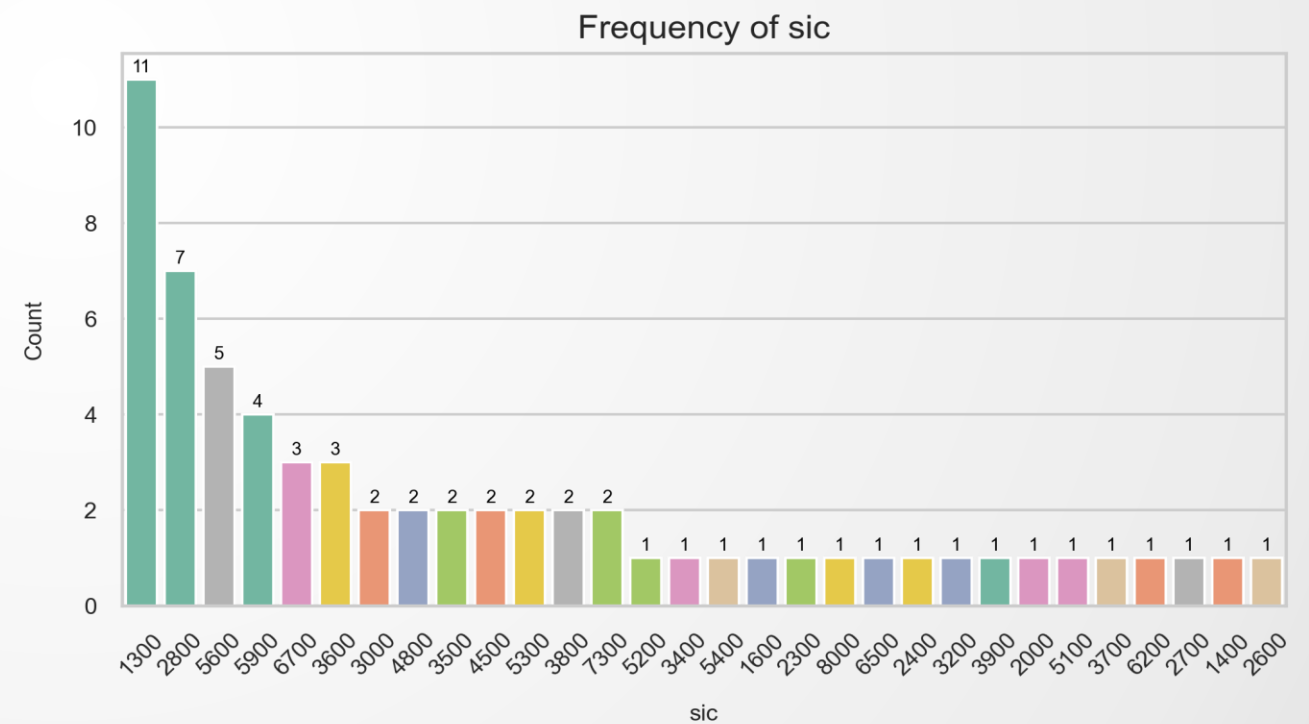
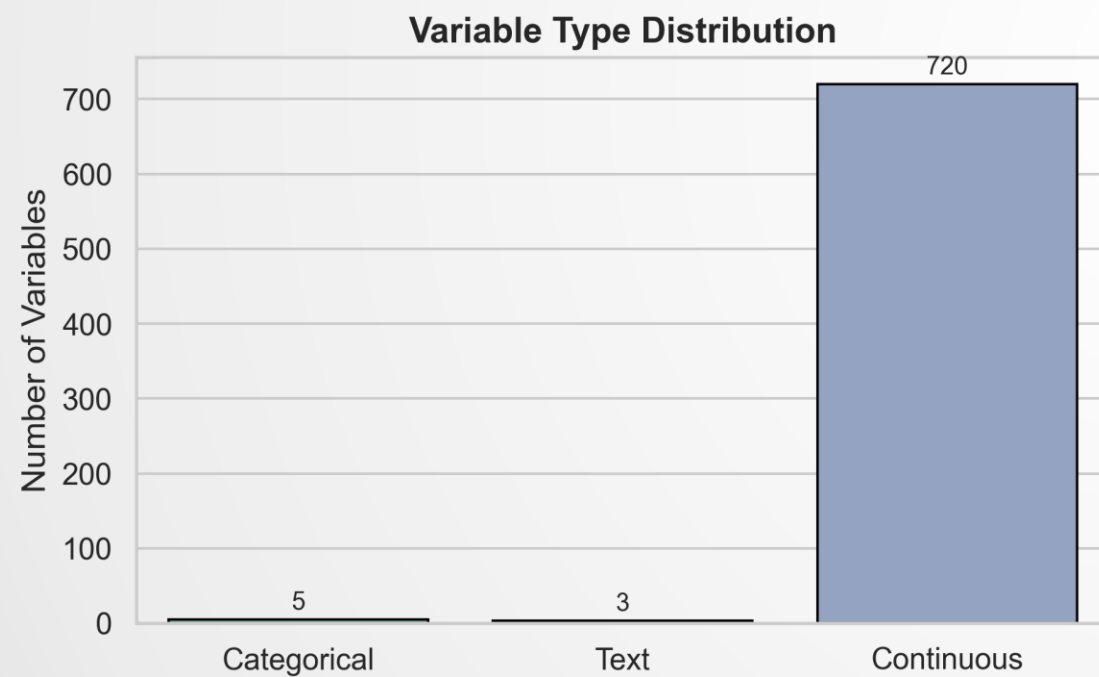
Motivation

- ▣ Corporate bankruptcy prediction
- ▣ High amount of financial features
- ▣ Class imbalance synthetic oversampling
- ▣ Feature selection (LASSO, XGBoost)
- ▣ Comparing Random Forest and SVM



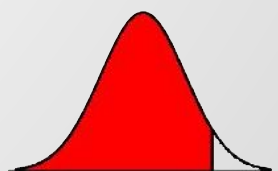
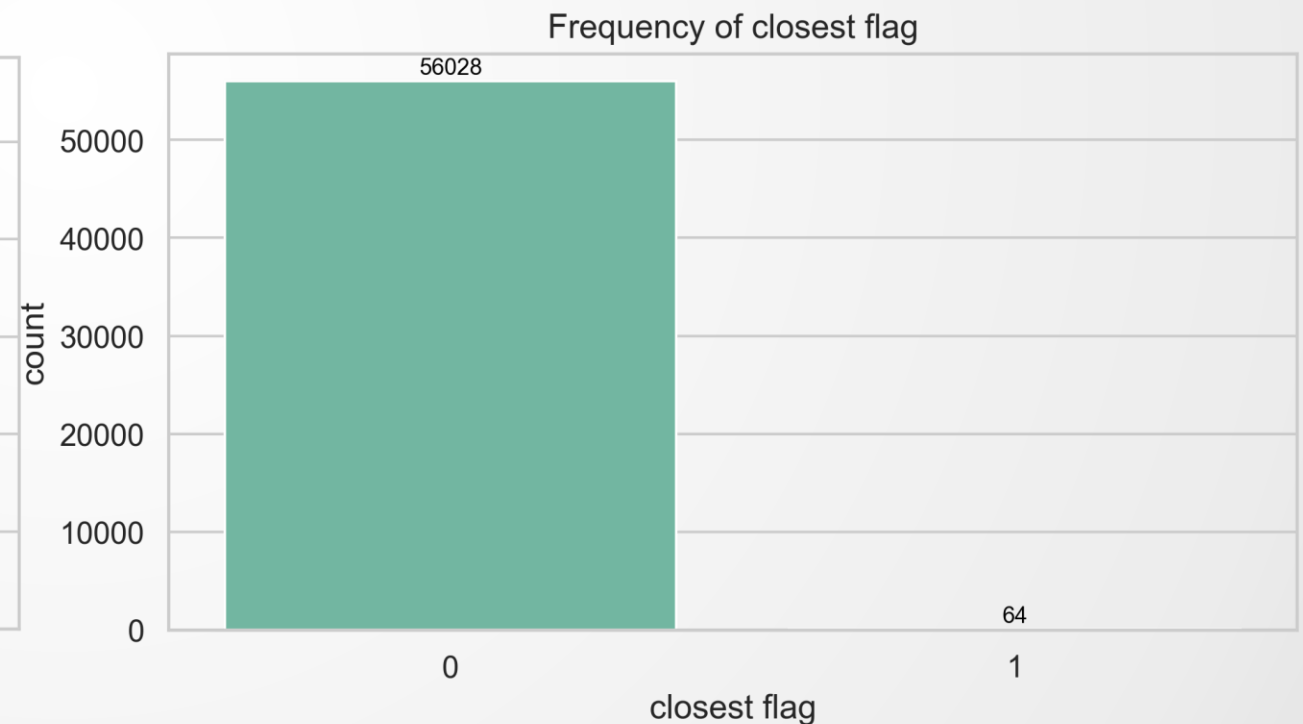
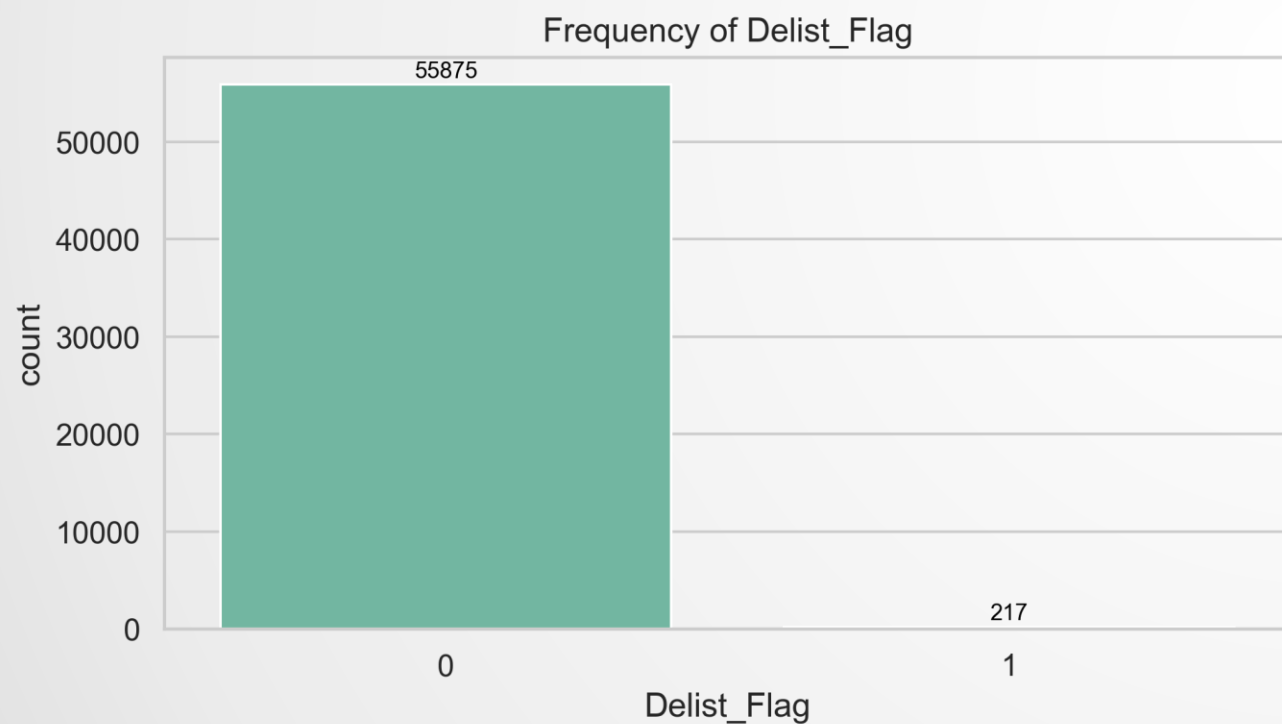
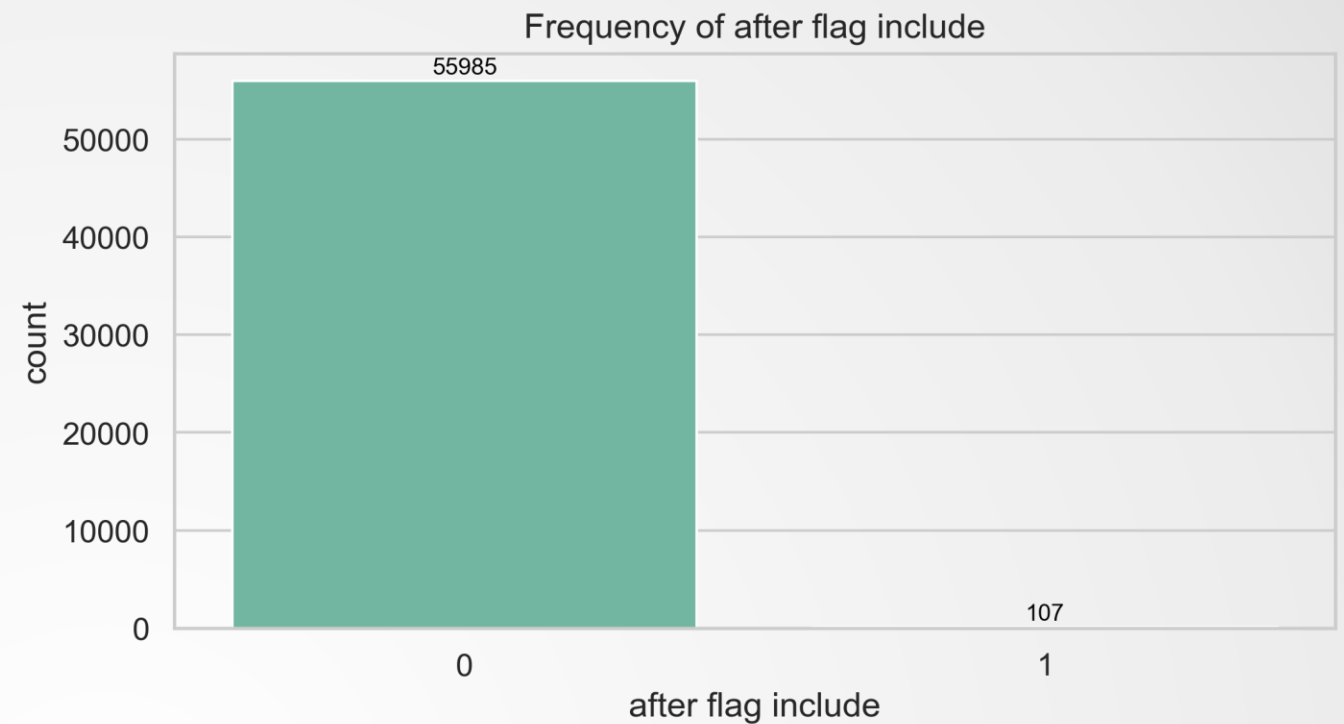
Data

- 11431 companies
- 728 features
- Large class imbalance



Data

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- 728 features
- Large class imbalance

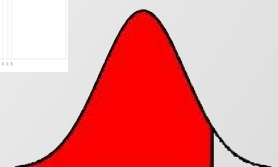
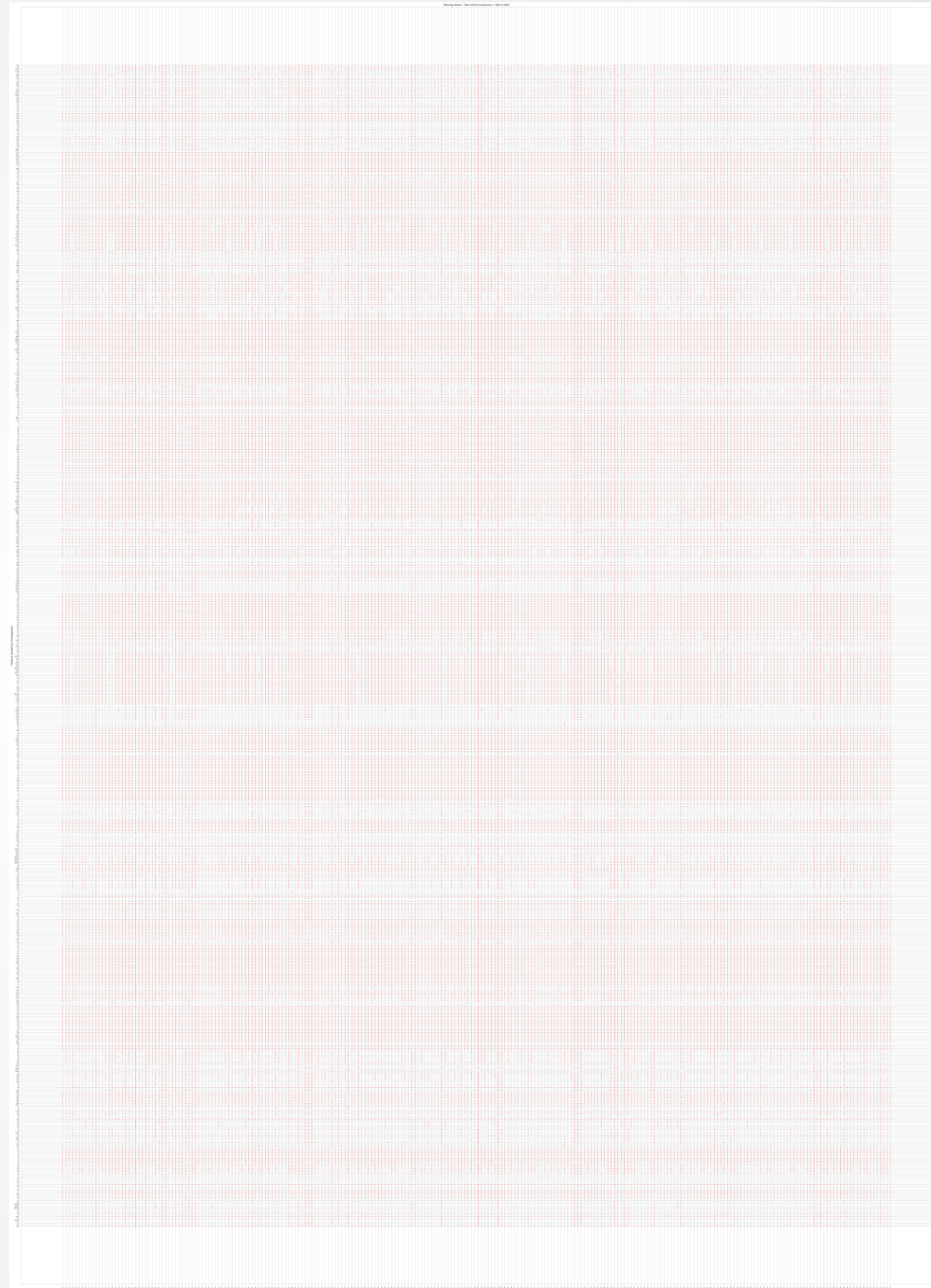


Data

- Many missing values
- Not evenly distributed
- Create missing indicator

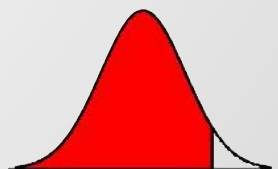
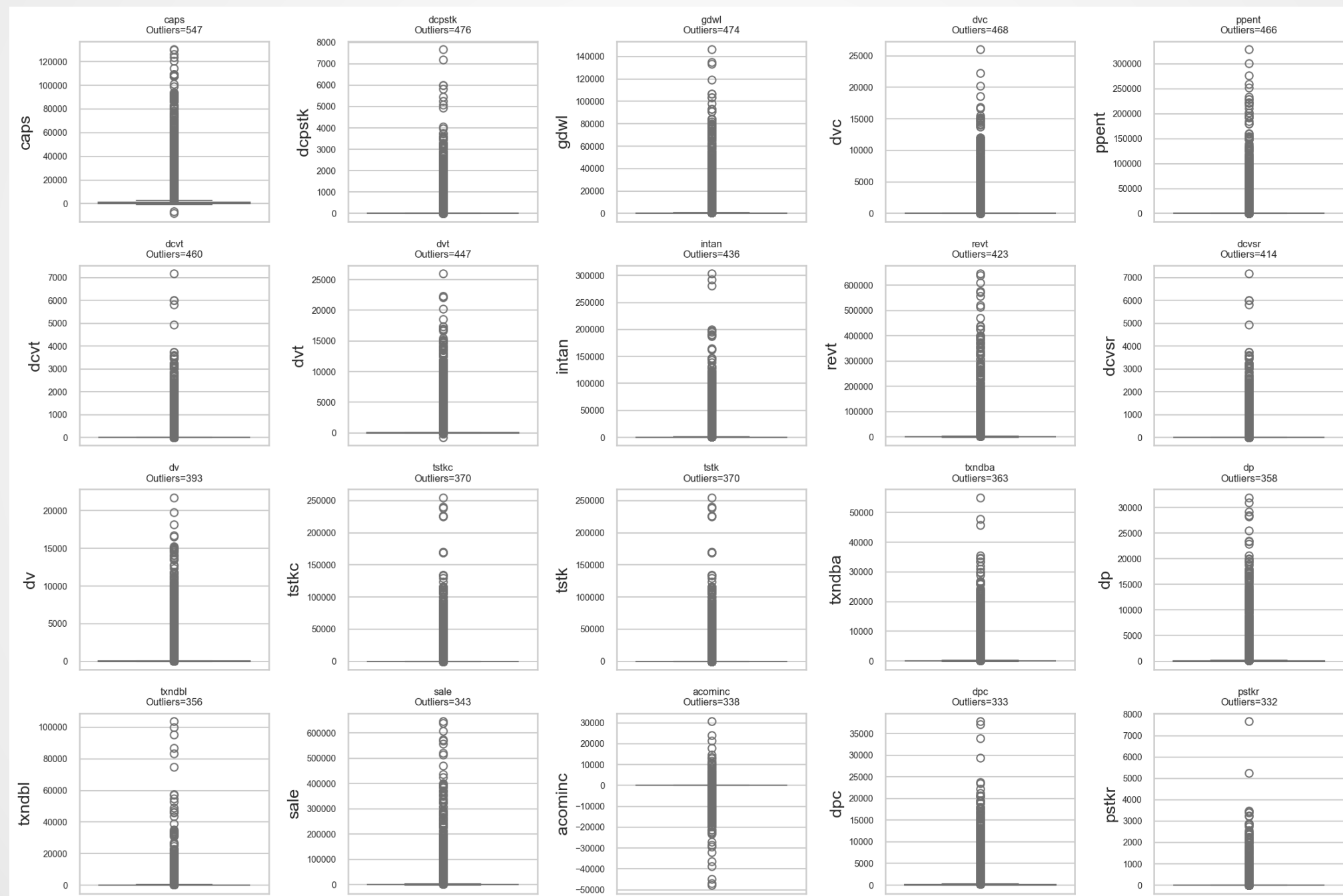
X axis: 250 companies, Y axis: 728 features

U.S. Corporate Bankruptcy Prediction



Data

Boxplots of select features

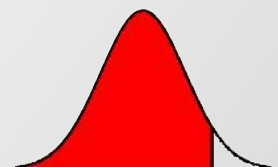


Data

- Description summary for selected numerical features
- Log transformation

	count	mean	std	min	25%	50%	75%	max	skewness	kurtosis
acco	3159	0.09	1.39	0.00	0.00	0.00	0.00	35.33	20.16	437.32
acdo	26555	2.91	111.17	0.00	0.00	0.00	0.00	11389.00	77.88	7108.72
aco	27745	216.30	2811.14	-25.00	0.57	6.49	43.41	197302.72	43.06	2259.30
acodo	26605	192.46	2855.48	-25.00	0.06	3.03	26.48	197223.17	42.82	2212.82
acominc	30837	-150.17	1235.43	-47852.00	-11.35	0.00	0.00	30738.00	-13.18	382.06
acox	26947	198.35	2843.88	-25.00	0.06	3.22	28.25	197223.17	42.81	2220.38
acoxar	246	1339.17	5677.29	0.00	8.76	44.00	363.03	60029.00	8.98	89.33
act	24567	1876.03	8706.38	0.00	27.50	206.39	861.25	203466.53	12.31	194.18
aedi	899	440.95	1428.51	0.00	0.99	24.06	186.69	11585.00	5.11	28.07
aldo	26592	3.06	132.55	0.00	0.00	0.00	0.00	17109.00	91.43	10784.25
ao	36251	1232.35	11414.53	-1.81	1.40	25.40	157.22	417337.00	18.24	397.28
aocidergl	29967	-1.47	130.69	-11935.00	0.00	0.00	0.00	2562.00	-41.28	3180.36
aociother	30331	-6.09	178.80	-11827.00	0.00	0.00	0.00	5414.00	-35.54	2173.08
aocipen	30094	-51.03	518.82	-24718.00	0.00	0.00	0.00	8924.00	-20.30	665.45
aocisecgl	30322	0	0	0	0	0	0	0	0	0

The entire description summary refers to overleaf table1



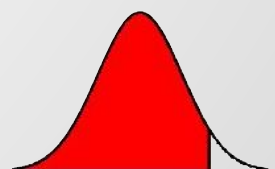
Benchmark

Dataset (period 2019-2024):

- ▣ USA annual financial statement
- ▣ USA Delisting dataset (Bankruptcy only)

Using financial statement to predict if it bankrupts.

- ▣ Training data: 2019-2023
- ▣ Testing data: 2024



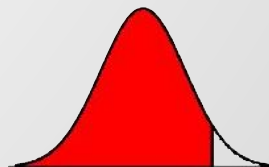
Benchmark

3 dependent variables:

```
merge_93420 = merge_del_flag.loc[(merge_del_flag['PERMNO'] == 93420)]
merge_93420
```

Year 2019-2024, total 6 years

vcapch	xidoc	PERMNO	Company Name	ticker	Delisting Date	Delisting_year	Delisting_quarter	Standard Industry Classification Code	Industry	CUSIP	closest flag	after flag include	Delist_Flag
NaN	0.000	93420.0	CHORD ENERGY CORP	CHRD	2020-10-09	2020.0	4.0	1311.0	OIL AND GAS EXTRACTION	674215207	0	0	1
NaN	0.000	93420.0	CHORD ENERGY CORP	CHRD	2020-10-09	2020.0	4.0	1311.0	OIL AND GAS EXTRACTION	674215207	1	1	1
NaN	166.338	93420.0	CHORD ENERGY CORP	CHRD	2020-10-09	2020.0	4.0	1311.0	OIL AND GAS EXTRACTION	674215207	0	1	1
NaN	428.007	93420.0	CHORD ENERGY CORP	CHRD	2020-10-09	2020.0	4.0	1311.0	OIL AND GAS EXTRACTION	674215207	0	1	1
NaN	0.000	93420.0	CHORD ENERGY CORP	CHRD	2020-10-09	2020.0	4.0	1311.0	OIL AND GAS EXTRACTION	674215207	0	1	1
NaN	0.000	93420.0	CHORD ENERGY CORP	CHRD	2020-10-09	2020.0	4.0	1311.0	OIL AND GAS EXTRACTION	674215207	0	1	1



Benchmark

Logistic regression

► how to fill up missing value? Mean v.s. Median

Dependence variable(y) = Closest flag

準確度: 0.9995670058454211

分類報告:

Mean

	precision	recall	f1-score	support
0	1.00	1.00	1.00	9236
1	0.00	0.00	0.00	2
accuracy			1.00	9238
macro avg	0.50	0.50	0.50	9238
weighted avg	1.00	1.00	1.00	9238

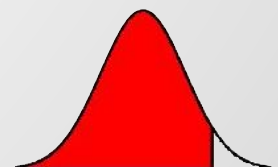
A bit worse.

準確度: 0.9991340116908421

分類報告:

Median

	precision	recall	f1-score	support
0	1.00	1.00	1.00	9236
1	0.00	0.00	0.00	2
accuracy			1.00	9238
macro avg	0.50	0.50	0.50	9238
weighted avg	1.00	1.00	1.00	9238



Benchmark

Logistic regression

► how to fill up missing value? Mean v.s. Median

Dependence variable(y) = After Flag Include

準確度: 0.9981597748430396

分類報告:

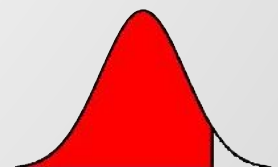
	precision	recall	f1-score	support
Mean				
0	1.00	1.00	1.00	9229
1	0.10	0.11	0.11	9
accuracy			1.00	9238
macro avg	0.55	0.56	0.55	9238
weighted avg	1.00	1.00	1.00	9238

A bit worse.

準確度: 0.9974020350725266

分類報告:

	precision	recall	f1-score	support
Median				
0	1.00	1.00	1.00	9229
1	0.00	0.00	0.00	9
accuracy			1.00	9238
macro avg	0.50	0.50	0.50	9238
weighted avg	1.00	1.00	1.00	9238



Benchmark

Logistic regression

► how to fill up missing value? Mean v.s. Median

Dependence variable(y) = Delist Flag

準確度: 0.9984845204589738

分類報告:

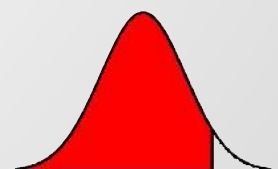
		precision	recall	f1-score	support
Mean	0	1.00	1.00	1.00	9229
	1	0.00	0.00	0.00	9
accuracy				1.00	9238
macro avg		0.50	0.50	0.50	9238
weighted avg		1.00	1.00	1.00	9238

A bit worse.

準確度: 0.9975102836111712

分類報告:

		precision	recall	f1-score	support
Median	0	1.00	1.00	1.00	9229
	1	0.00	0.00	0.00	9
accuracy				1.00	9238
macro avg		0.50	0.50	0.50	9238
weighted avg		1.00	1.00	1.00	9238



Benchmark

□ Logistic regression

▶ adding the missing value indicator

Dependence variable(y) = Closest flag

準確度: 0.9995670058454211

分類報告:

Mean		precision	recall	f1-score	support
	0	1.00	1.00	1.00	9236
	1	0.00	0.00	0.00	2
	accuracy			1.00	9238
	macro avg	0.50	0.50	0.50	9238
	weighted avg	1.00	1.00	1.00	9238

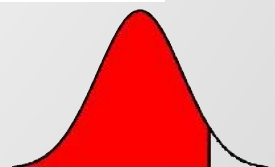
A bit worse.



準確度: 0.9994587573067764

分類報告:

		precision	recall	f1-score	support
	0	1.00	1.00	1.00	9236
	1	0.00	0.00	0.00	2
	accuracy			1.00	9238
	macro avg	0.50	0.50	0.50	9238
	weighted avg	1.00	1.00	1.00	9238



Benchmark

Logistic regression

adding the missing value indicator

Dependence variable(y) = After Flag Include

準確度: 0.9981597748430396

分類報告:

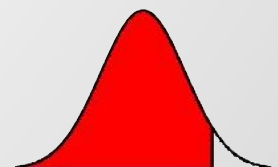
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Mean				
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1	0.10	0.11	0.11	9
accuracy			1.00	9238
macro avg	0.55	0.56	0.55	9238
weighted avg	1.00	1.00	1.00	9238

A bit better.

準確度: 0.9988092660749079

分類報告:

	precision	recall	f1-score	support
0	1.00	1.00	1.00	9229
1	0.25	0.11	0.15	9
accuracy			1.00	9238
macro avg	0.62	0.56	0.58	9238
weighted avg	1.00	1.00	1.00	9238



Benchmark

Logistic regression

adding the missing value indicator

Dependence variable(y) = Delist Flag

準確度: 0.9995670058454211

分類報告:

Mean

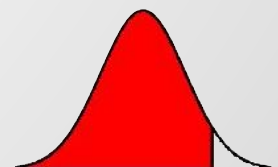
	precision	recall	f1-score	support
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1	0.00	0.00	0.00	2
accuracy			1.00	9238
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weighted avg	1.00	1.00	1.00	9238

A bit worse.

準確度: 0.9985927689976185

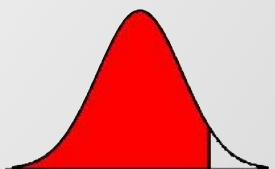
分類報告:

	precision	recall	f1-score	support
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1	0.00	0.00	0.00	9
accuracy			1.00	9238
macro avg	0.50	0.50	0.50	9238
weighted avg	1.00	1.00	1.00	9238



Benchmark

- Data Imbalance
- Much Less Samples of $y=1$
 - ▶ add more datapoints by including longer period





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